

Developing Applications Using NoSQL Databases



NoSQL (or 'Not only SQL') databases offer the means to store and retrieve data that is not stored in the traditional RDBMS style of tabular columns. NoSQL is rapidly finding a place in Big Data and realtime Web applications.

For the past few years, the world of Web technologies has been associated with assorted programming languages and scripts. The domain of Web development is not confined to a specific programming language or library, using which Web applications and portals are developed. Thousands of toolkits, programming paradigms, scripts, databases and application programming interfaces (APIs) are in use for multiple services and applications. The days when only a static website was needed using classical hypertext markup language (HTML) are long gone. The corporate world is using different tools to design, develop and launch applications with maximum user interactions as well as effective graphical user interfaces (GUI).

SQL, NewSQL and NoSQL databases

Whenever there is need of a Web 2.0 portal (an interactive website), database-oriented applications are required, as the back-end database keeps and maintains the records required for the appropriate functioning of the modules. For example, guest book messages, posts, blogs, e-mail messages, chat messages and comments are stored in the back-end databases so that these can be retrieved, processed or edited at any instant.

Classically, the RDBMS packages used for database applications include MySQL, Oracle, Apache Derby, IBM DB2, IBM Notes, Microsoft SQL Server, PostgreSQL, SQLite, Sybase and many others. These are known as the traditional